Tutorial: Neural Networks

Multi-output Neural Network, Backpropagation Learning

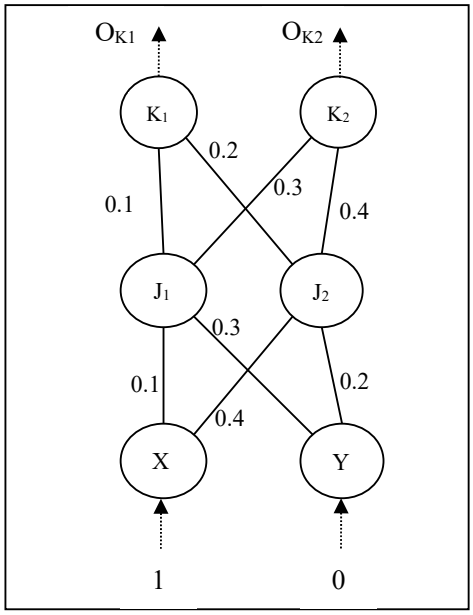


Fig.1 Initial Network Configuration

Parameters for Backpropagation learning:
Learning rate η = 2
No bias nodes are used.
The activation function used is the sigmoid function.

Training set

Pattern#	Input: **x**	Input: **y**	Target output: T_{K1}	Target output: T_{K2}
1	1	1	1	0
2	1	0	0	1
3	0	1	1	1
…	…	…	…	…

Network Evaluation

1. Given the initial network configuration in Fig. 1 and the parameters for Backpropagation learning, put one of the training patterns to be learned (training pattern #2) on the input units, and then calculate the actual outputs of the network (i.e. O_{K1} and O_{K2}).

Minimisation of Error

2. Based on the results of your calculations in Item#1, calculate the error signals at the output nodes (i.e. δ_{k1} and δ_{k2}).

3. Backpropagate δ_{k1} and δ_{k2}, then calculate the weight updates for W_{j1k1}, W_{j1k2} W_{j2k1} and W_{j2k2}

4. Calculate the error signal at the hidden nodes j1 and j2 (i.e. δ_{j1} and δ_{j2}).

5. Update the weights W_{xj1}, W_{xj2}, W_{yj1} and W_{yj2}

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